

WHAT WE FOUND

Compressing a judgment to its gist is a **double-edged prior**: it **wins when labels are scarce**, but a plain hierarchical model **overtakes it once labels are plentiful**.

N = 10

GlocalDoc **0.044** vs H-MLM 0.019 → **2.3× macro-F1**, and far more stable (H-MLM fails on 2/5 seeds)

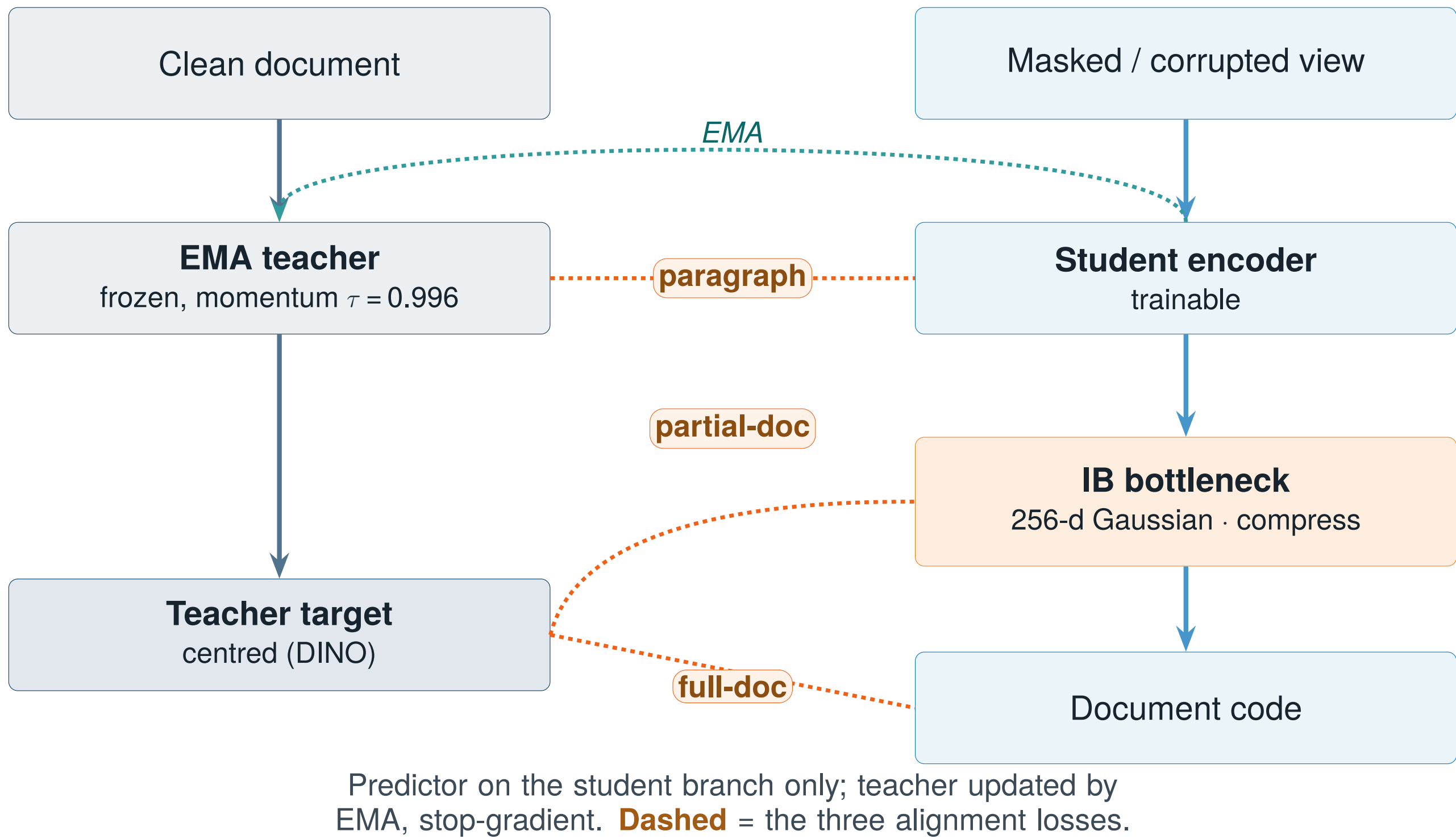
N ≥ 50

H-MLM **0.273** vs GlocalDoc 0.108 → **the ordering flips** as labels grow

1 What we do & why

- **The problem.** Language models read sentences well but lose the document as a *whole*; in a legal judgment the decisive signal is spread across many paragraphs, and expert labels are scarce (**few-shot**).
- **What we build.** A frozen **teacher** reads the clean judgment; a **student** reads a masked, partial copy and must reproduce the teacher's reading at **three scales** — paragraph, partial- and whole-document — squeezed through a compressive **information bottleneck**.
- **The roots.** It adapts the *Glocal Information Bottleneck* (Yang 2025) from time series to legal text, and is tested against hierarchical MLM (**H-MLM**) and a no-pretrain baseline on **ECTHR** judgments (LexGLUE; 10 labels, $\approx 100\times$ imbalance, macro-F1).
- **We ask three questions.**
 - RQ1 Does GlocalDoc beat a hierarchical MLM at few-shot?
 - RQ2 Does the advantage depend on the label budget N ?
 - RQ3 What breaks on boilerplate, and what fixes it?

2 The architecture



3 Preventing collapse

- The failure.** ECTHR judgments are mostly boilerplate; the raw encoder sees documents as *nearly identical* (median pairwise cosine **0.9997**). The teacher target becomes constant and the student **collapses**.
- The fix.** **DINO centring** subtracts a running mean from the teacher target. Inter-document cosine of the student code falls **0.42 to <0.05** and stays stable: the single change that keeps training stable.
- Also.** EMA teacher ($\tau = 0.996$), predictor on the student only, VICReg variance & covariance.

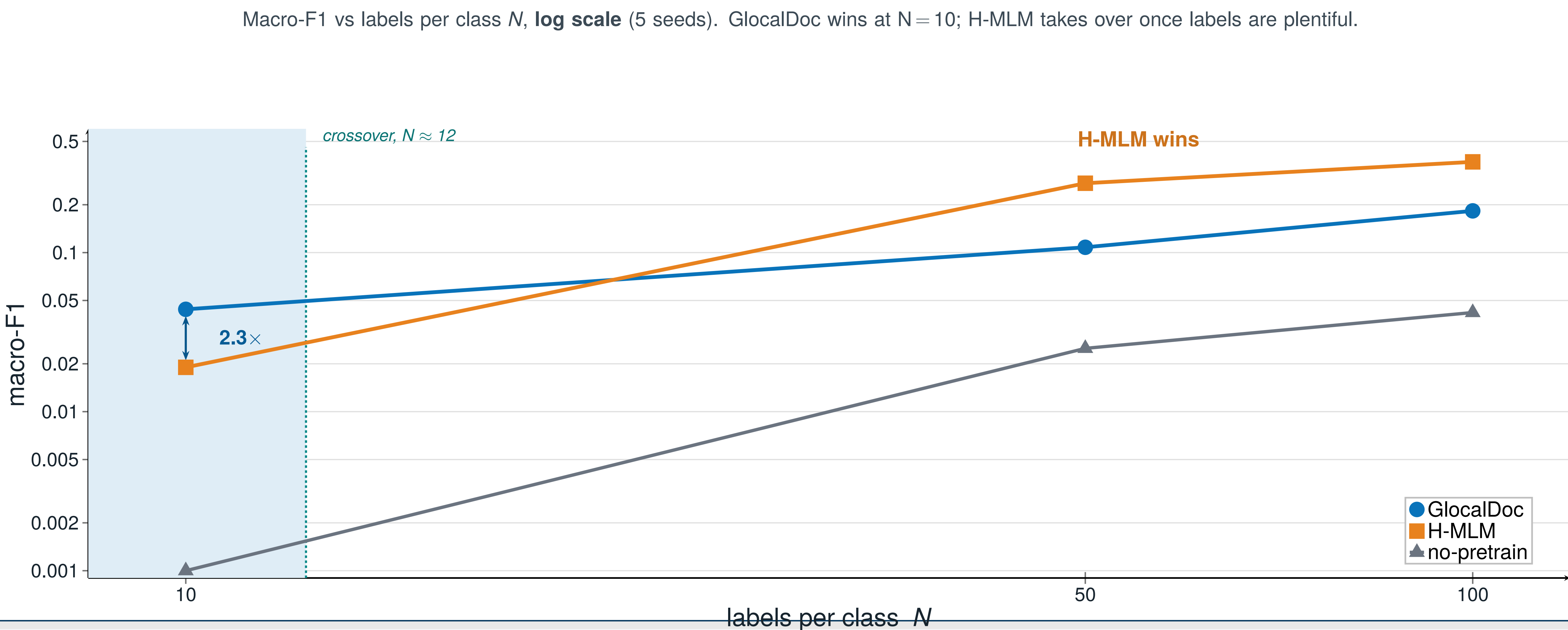
4 Results

Macro-F1, mean $\pm$ sd over 5 seeds

Pre-training	N=10	N=50	N=100
GlocalDoc	0.044 $\pm$.009	0.108 $\pm$.024	0.183 $\pm$.006
H-MLM	0.019 $\pm$.023	0.273 $\pm$.061	0.372 $\pm$.032
no-pretrain	0.001	0.025	0.042

Best score per column in **bold**. At $N = 10$, **2 of 5** H-MLM seeds collapse to ≈ 0 while all 5 GlocalDoc seeds stay in **0.034 to 0.057**: stability, not just a higher mean.

5 The crossover



Bottom line: the bottleneck is a strong prior — a lifesaver under extreme scarcity, a handicap once labels are plentiful (it discards the lexical detail the MLM keeps). **Match the objective to the data regime.**